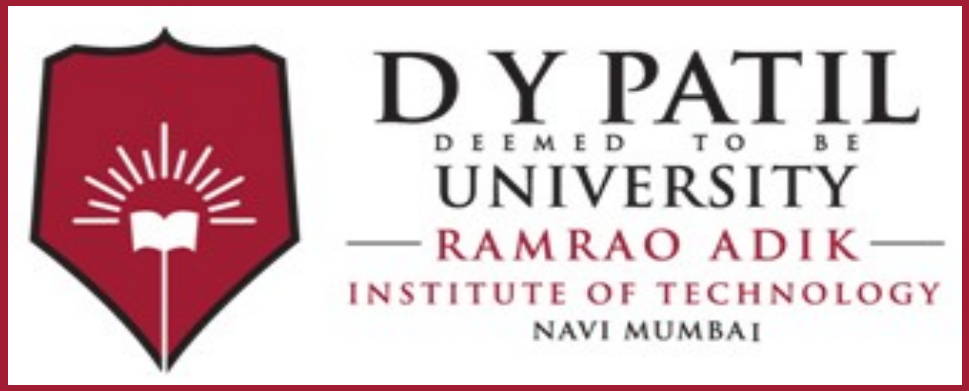


Web Development and Analytics
Unit 4: Digital forensic

Faculty Name : Ms. Priyanka Shingane

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Lecture 1

Digital forensic



Introduction to Digital Forensics

•Digital Forensics is the process of identifying, preserving, analyzing, and presenting digital evidence in a legally acceptable manner.

Purpose:

•Investigate cybercrimes.

•Support legal proceedings.

•Recover lost/deleted information.

Branches of Digital Forensics:

•**Computer Forensics** – PCs, laptops, servers.

•**Network Forensics** – logs, intrusion detection, traffic capture.

•**Mobile Forensics** – smartphones, SIM cards, apps.

•**Database Forensics** – structured storage systems.

•**Cloud Forensics** – distributed, virtualized environments.

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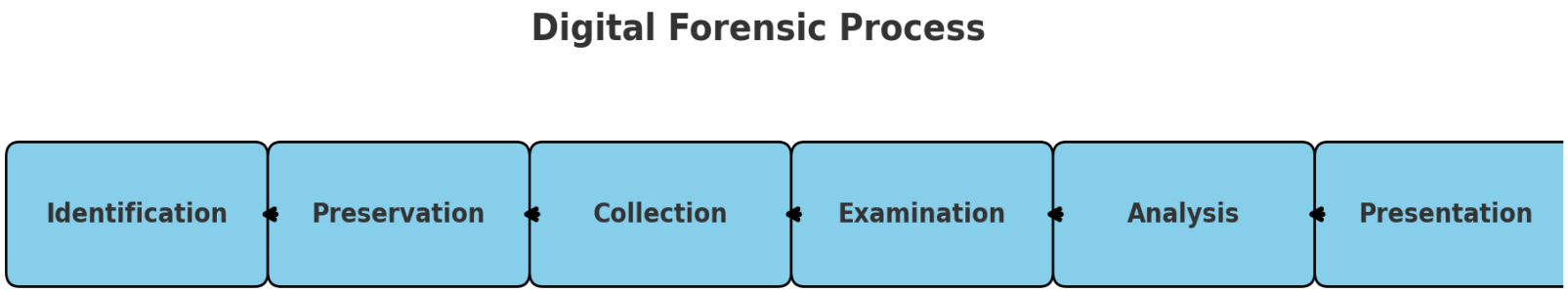
Digital Evidence

- **Types:**
  - Volatile (RAM, cache, active processes).
  - Non-volatile (hard drives, flash drives, logs).
- **Characteristics:**
  - Easily altered.
  - Requires proper handling.
- **Legal Admissibility Requirements (Daubert Standards):**
  - Reliability.
  - Relevance.
  - Integrity maintained through chain of custody.

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Digital Forensic Process (Also known as **The Forensic Lifecycle**)

1. **Identification** – Detecting potential evidence (devices, logs).
2. **Preservation** – Ensuring data is not modified (write blockers, imaging).
3. **Collection/Acquisition** – Creating bit-by-bit copies (forensic imaging).
4. **Examination** – Filtering, recovering deleted files, keyword searches.
5. **Analysis** – Reconstructing events, timeline analysis, correlation.
6. **Presentation** – Preparing reports for court.



Tools & Techniques

- **Forensic Tools:**
  - EnCase, FTK (Forensic Toolkit), Autopsy/Sleuth Kit.
  - Wireshark (network), Cellebrite (mobile).
- **Techniques:**
  - Hashing (MD5, SHA-1) → verify data integrity.
  - Timeline analysis.
  - Log analysis.
  - File carving.
  - Registry and memory analysis.

Chain of Custody

Documentation of evidence handling from collection to court.

- **Steps:**
  - Who collected the evidence.
  - When & where it was collected.
  - How it was stored and transferred.
- **Importance:** Ensures evidence is **legally admissible**.

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Challenges in Digital Forensics

- Huge volume of data.
- Encryption & steganography.
- Cloud computing & jurisdiction issues.
- Anti-forensics techniques (data wiping, obfuscation).
- Maintaining privacy & ethical concerns.

Applications of Digital Forensics

- Cybercrime investigation (hacking, phishing, fraud).
- Corporate investigations (IP theft, insider threats).
- Civil cases (employment disputes, harassment).
- Incident response (malware, ransomware attacks).

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Cybercrime Scene Investigation (CCSI)

- Cybercrime Scene Investigation = Applying forensic principles to investigate crimes in a **digital environment**.
- Goal: Identify, collect, preserve, and analyze digital evidence in a manner acceptable in court.
- Different from physical crime scenes → requires both **digital forensics** and **traditional investigation**.

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Types of Cybercrime Scenes

1.Physical Crime Scene with Digital Devices

1. PCs, laptops, USB drives, mobile phones, routers, CCTV DVRs.

2.Virtual/Digital Crime Scene

1. Cloud storage, online accounts, social networks, virtual machines, remote servers.

3.Hybrid Scene

1. Combination of physical + online evidence.

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Cybercrime Scene Investigation Process

A. Preparation

Secure legal authorization (search warrant).
Team briefing (roles, responsibilities).
Prepare forensic toolkit (write blockers, imaging tools, evidence bags).

B. Securing the Scene

Isolate devices (prevent remote wiping/tampering).
Disable network connections carefully (pull network cable, not power).
Photograph the scene and label evidence.

C. Evidence Collection

Physical: seize computers, storage media, smartphones.
Volatile data: capture RAM, running processes, active connections.
Non-volatile: disk imaging, logs, emails, chat records, deleted files.

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D. Preservation

- Use **forensic imaging** (bit-by-bit copy).
- Apply **hashing (MD5/SHA)** to ensure integrity.
- Maintain **chain of custody**.

E. Examination & Analysis

- Recover deleted files, analyze logs, emails, chat history.
- Timeline reconstruction.
- Malware/Rootkit detection.
- Correlation of activities across devices/networks.

F. Reporting & Presentation

- Write clear, structured report.
- Present evidence in court with explanations.
- Ensure compliance with laws and standards.

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Tools Used

- **Forensic Acquisition**: EnCase, FTK Imager, Autopsy, X-Ways.
- **Memory Forensics**: Volatility, Rekall.
- **Network Forensics**: Wireshark, tcpdump.
- **Mobile Forensics**: Cellebrite, Oxygen Forensics.

Challenges

- **Encryption & Steganography**.
- **Jurisdiction issues** (servers in other countries).
- **Cloud forensics** (limited access to physical hardware).
- **Anti-forensics** (data wiping, obfuscation, time manipulation).

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Cybercrime Scene Investigation: Identifying and Documenting the Scene & Evidence

Importance

- First and most critical stage in cybercrime scene investigation.
- Ensures **no evidence is overlooked** and provides a **baseline for integrity**.
- Poor identification/documentation may cause evidence to be **inadmissible in court**.

Identification of the Scene

- **Physical Scene**:
  - Computers, laptops, mobile devices, USB drives, external hard disks.
  - Networking equipment: routers, switches, firewalls.
  - Peripheral devices: printers, cameras, DVRs.
- **Digital/Virtual Scene**:
  - Cloud storage, email accounts, social media, online banking.
  - Server logs, databases, virtual machines, IoT devices.
- **Hybrid Scene**:
  - Combination of both physical & digital (e.g., laptop + cloud accounts).

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Identification of Evidence

- **Volatile Evidence (short-lived)**:
  - RAM contents, cache memory, system processes, network connections.
  - Must be collected **immediately** (before power-off).
- **Non-Volatile Evidence**:
  - Hard disks, USB drives, log files, emails, cloud storage.
  - Survives system shutdown.

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Documentation Process

- **Visual Documentation**:
  - Take photographs of devices, screen displays, network cables, and device placement.
  - Screenshots of active windows or messages.
- **Written Documentation**:
  - Device details: make, model, serial numbers.
  - Description of scene environment (time, date, location, who was present).
  - Record device state: powered on/off, connected/disconnected.
- **Labeling & Tagging**:
  - Each piece of evidence should be assigned a unique ID.
  - Evidence sealed in proper bags/containers with label tags.

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Example Documentation Table

Evidence ID	Device Type	Serial Number	Condition	Location Found	Notes
E001	Laptop	SN12345	Powered On	Desk in office	Logged-in user session active
E002	USB Drive	SN56789	Sealed	Drawer	32GB Kingston, labeled “work”
E003	Router	SN78901	Connected	Server room	Active connections observed

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Maintaining the Chain of Custody

- **Chain of Custody (CoC)** is the **chronological documentation** of how digital evidence is collected, handled, stored, and transferred.
- Ensures that evidence remains **authentic, reliable, and admissible in court**.
- Proves that the evidence presented in court is the **same as originally collected**.
- Prevents tampering, contamination, or unauthorized access.
- Protects the credibility of forensic investigators.

Key Principles

1. **Evidence Integrity** – must remain unaltered.
2. **Proper Documentation** – every step must be recorded.
3. **Authorized Access Only** – only certified/authorized investigators can handle.
4. **Secure Storage** – physical and digital safeguards.

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Chain of Custody Process

A. Collection

Evidence is identified, photographed, labeled, and packaged.

Use **evidence tags** (with unique ID numbers).

Seize both **physical devices** (laptop, mobile) and **digital data** (logs, emails).

B. Preservation

Store in tamper-proof bags/containers.

Use **write-blockers** while imaging drives.

Generate **hash values** (MD5/SHA-256) to confirm integrity.

C. Documentation

Record details in a **Chain of Custody form**:

- Evidence ID, description, serial number.
- Who collected the evidence.
- Time, date, and location of collection.
- Purpose of transfer (e.g., to lab, to court).

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D. Transfer

When evidence changes hands, both **giver and receiver sign**.

Each transfer must be logged with **date, time, and signatures**.

E. Storage

Keep in **secure evidence lockers** or forensic labs.

Access logs maintained to track entry/exit

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Example: Chain of Custody Form

Evidence ID	Description	Collected By	Date/Time	Transferred To	Purpose	Signature
E001	Laptop (Dell, SN12345)	Officer A	10-09-2025, 11:30 AM	Forensic Lab	Imaging	A. Officer
E001	Laptop Clone Image	Analyst B	10-09-2025, 2:00 PM	Analyst C	Malware Analysis	B. Analyst

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Best Practices

- Always seal evidence with tamper-evident labels.
- Use **digital logs + physical paperwork** for redundancy.
- Restrict access (only designated officers/analysts).
- Hash verification after every transfer.
- Maintain records **until case closure**.

Challenges

- **Cloud-based data** (multiple jurisdictions).
- **Encryption** may require multiple handling attempts.
- **Improper documentation** may lead to evidence rejection in court

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Cloning of Evidence (Forensic Imaging)

- **Cloning of evidence** (also called **forensic imaging**) is the process of making an **exact bit-by-bit copy** of digital storage media (hard drives, SSDs, USBs, mobile storage) for forensic analysis.

- It includes **all files, deleted data, slack space, hidden partitions** – not just visible files.

Purpose

To ensure the **original evidence is never altered**.

Investigators work only on the **clone (forensic image)**.

Protects the **integrity and admissibility** of evidence in court.

Characteristics of a Forensic Clone

Bit-by-bit identical copy of the original media.

Includes both **allocated and unallocated space**.

Hash verification (MD5/SHA-256) ensures integrity.

Created with **write-blockers** to prevent modification.

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Steps in Cloning

1. **Preparation**
    1. Use forensic workstation.
    2. Connect storage via **hardware write-blocker**.
2. **Hash Original Media**
    1. Generate hash value before cloning.
    2. Example: SHA-256 fingerprint.
3. **Create Clone (Forensic Image)**
    1. Use forensic tools:
        1. *FTK Imager, EnCase, X-Ways Forensics, dd (Linux), Guymager*.
4. **Hash Verification**
    1. Compute hash of cloned image.
    2. Must match original device hash.
5. **Secure Storage**
    1. Store original in evidence locker.
    2. Work only on forensic image.

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Tools for Cloning

- **FTK Imager** – GUI-based imaging & verification.
- **EnCase** – commercial forensic suite.
- **dd (Linux command)** – creates raw bit-by-bit copies.
- **Guymager** – open-source forensic imager.
- **Cellebrite UFED** – for mobile device cloning.

Best Practices

- Never analyze the original device.
- Use **write-blockers** to prevent accidental modification.
- Always generate and store **hash values** for verification.
- Document cloning process in **Chain of Custody forms**.

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Live and Dead System Forensics

- In **cybercrime scene investigation**, evidence can be gathered from both **active (running) systems** and **inactive (powered-off) systems**.

- These two approaches are called:
- **Live System Forensics**
- **Dead System Forensics**

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Aspect	Live System Forensics	Dead System Forensics
Definition	Examination of a running system to capture volatile data before it is lost.	Examination of a powered-off system by analyzing a forensic image (clone).
Data Collected	- RAM contents (processes, encryption keys, clipboard data) - Running applications & services - Network connections & active sessions - Logged-in users & passwords - System date & time	- Hard disk content (all files, deleted files, slack space) - File systems & directories - Log files & registry data - Hidden/encrypted partitions - Persistent malware
Tools Used	- Volatility Framework (RAM analysis) - DumpIt (memory capture) - FTK Imager (Live) - Wireshark / tcpdump (network capture)	- EnCase - Autopsy - Sleuth Kit - X-Ways Forensics - dd / Guymager (disk imaging)

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Advantages	- Access to volatile data not available after shutdown - Detects live attacks (malware, rootkits, ransomware) - Helps in insider threat investigations	- Preserves original system state - Reliable for long-term analysis - Safer for chain of custody
Challenges	- Risk of altering evidence during interaction - Requires high expertise - Time-sensitive, as volatile data changes quickly	- Cannot capture volatile memory (RAM, network sessions) - Encrypted files may be inaccessible without keys - Some evidence lost once powered off

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Perseveration of Evidence

- Preserved evidence ensures **credibility** of investigation.
- Protects investigators and organizations from **legal challenges**.
- Makes digital evidence **reliable and acceptable in court**

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Techniques and Methods

1.Forensic Imaging / Cloning

- Create an exact **bit-by-bit copy** of storage devices (HDD, SSD, USB, memory cards).
- Tools: dd, FTK Imager, Guymager, EnCase.
- Original device is sealed and preserved; investigation is performed on the clone.

2.Write Blockers

- Hardware or software tools that allow data to be read from a device **without modifying it**.
- Ensures **no accidental changes** to the original evidence.

3.Hashing (Integrity Verification)

- Use cryptographic hash algorithms (MD5, SHA-1, SHA-256) to generate a **digital fingerprint** of evidence.
- After copying, hashes of the original and copy are compared → if identical, integrity is preserved.

4.Secure Storage and Documentation

- Evidence stored in **tamper-proof containers** or evidence bags.
- Strict **access control** (only authorized personnel).
- Environmental protection (against dust, heat, magnetic fields).

5.Chain of Custody Forms

- Every transfer of evidence is recorded: who handled it, when, why, and how.
- Prevents allegations of tampering.

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Hashing Concepts to Maintain Integrity of Evidence

- **Hashing** is the process of generating a **fixed-length string (hash value or digest)** from input data of any size using mathematical algorithms.
- In digital forensics, hashing is used to **prove that evidence has not been altered** from the time of collection until it is presented in court.

Importance of Hashing in Forensics

- Ensures **data integrity** → even a **1-bit change** in evidence changes the hash drastically.
- Provides a **digital fingerprint** of evidence.
- Used to verify **forensic images/clones** against the original media.
- Accepted by courts as a **standard method of validation**.

- Hashing is a **one-way function** → you cannot reconstruct original data from hash.
- Even the smallest alteration (like changing one letter in a file) results in a **completely different hash**.
- Courts accept hash values as **mathematical proof of authenticity**.

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Common Hashing Algorithms Used

1.MD5 (Message Digest 5)

- 128-bit hash value (32-character hexadecimal).
- Example: d41d8cd98f00b204e9800998ecf8427e
- Fast but weaker (collision possible).

2.SHA-1 (Secure Hash Algorithm 1)

- 160-bit hash value (40 characters).
- Example: da39a3ee5e6b4b0d3255bfef95601890afd80709
- More secure than MD5 but also prone to collisions (deprecated in many cases).

3.SHA-256 (Secure Hash Algorithm 256-bit)

- 256-bit hash value (64 characters).
- Example: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
- Widely used today for forensic integrity.

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Process of Using Hashing in Evidence Handling

1. **Before Imaging (Original Evidence)**
   1. Compute the **hash of the original evidence** (disk, USB, file, etc.).
   2. Record it in the evidence log.
2. **After Imaging (Forensic Copy)**
   1. Create a forensic **bit-by-bit clone** using imaging tools.
   2. Compute the hash of the copy.
3. **Verification**
   1. Compare the **original hash vs cloned hash**.
   2. If identical → evidence integrity preserved.
4. **During Analysis**
   1. Any file extracted or analyzed can be hashed again to ensure no modification.
5. **Before Court Presentation**
   1. Investigator provides **hash values in the report** as proof of integrity.

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Examination and Analysis of Evidence

- Once evidence has been **identified, preserved, cloned, and hashed**, the next crucial steps are **examination** and **analysis**.
- These steps help investigators **extract useful information**, **establish patterns**, and **build a case** for legal proceedings.

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Examination of Evidence

Definition

The process of **systematically inspecting and extracting data** from collected digital evidence **without altering the original source**.

Focuses on identifying, recovering, and categorizing relevant information.

Tasks Involved

1. **File System Examination**

   Review of directories, hidden files, deleted files, slack space, unallocated space.

   Tools: Autopsy, Sleuth Kit, EnCase.

2. **Metadata Extraction**

   Document properties, timestamps (MAC times: Modified, Accessed, Created).

   Helps in timeline reconstruction.

3. **Keyword and String Searches**

   Search for suspicious terms (passwords, IPs, emails, malware signatures).

4. **Data Recovery**

   Retrieve deleted, formatted, or hidden data.

5. **Malware and Log Analysis**

   Detect malicious executables, registry modifications, abnormal logins, failed attempts.

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Analysis of Evidence

Definition

The process of **interpreting the extracted evidence** to determine **how the crime occurred, who was involved, and what damage was caused**.

Tasks Involved

1. Timeline Analysis

Correlate events (file modifications, logins, network connections).

Reconstruct attacker activity.

2. Correlating Multiple Sources

Match data from devices, networks, servers, and cloud logs.

Example: Matching an email's header with firewall logs.

3. Attribution (Linking Suspects to Crime)

Linking IP addresses, MAC addresses, user accounts, or recovered files to specific individuals.

4. Behavioral Analysis

Identify attacker's techniques (phishing, brute force, malware injection, insider threat).

5. Validation

Re-verify findings using hashing to confirm no evidence was altered during examination.

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Tools Used

- **Examination**: Autopsy, EnCase, FTK, X-Ways Forensics.
- **Analysis**: Volatility (memory), Wireshark (network), Splunk (logs), Maltego (link analysis).

Challenges

- Large volume of data (terabytes of logs/files).
- Strong encryption and password protection.
- Anti-forensic techniques (file wiping, steganography).
- Time constraints for reporting to court.

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Report Generation in Digital Forensics

Objectives of Report Generation

- Summarize the **process, tools, and methods** used.
- Present **facts and findings** in a clear, unbiased manner.
- Ensure the report is **understandable by both technical and non-technical readers** (lawyers, judges, jury).
- Provide **evidence of integrity** (hash values, chain of custody).
- Support legal decision-making.

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Key Components of a Forensic Report

1. Cover Page

Title of report

Case number, investigator's name, date

2. Executive Summary

High-level overview of the case and key findings.

3. Objectives & Scope

Purpose of the investigation (e.g., data breach, fraud, insider attack).

Limitations (e.g., encrypted files not accessed).

4. Methodology

Steps followed: Identification → Preservation → Collection → Examination → Analysis.

Tools used (FTK, EnCase, Volatility, Wireshark, etc.).

Imaging and hashing details.

5. Evidence Details

Description of seized devices (serial numbers, type, size).

Chain of custody documentation.

Hash values of original and cloned media.

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6. Examination Findings

•Files recovered (deleted/hidden files).

•Metadata (timestamps, user info).

•Network logs, registry entries, memory captures.

7. Analysis Results

•Timeline reconstruction of the incident.

•Correlation of multiple data sources.

•Attribution to suspects/users.

8. Conclusion

•Final interpretation of results.

•Whether evidence supports/disproves allegations.

9. Appendices

•Screenshots of findings.

•Hash value tables.

•Tool logs.

•Glossary of technical terms.

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Qualities of a Good Forensic Report

Accuracy – no assumptions or guesses.

Clarity – simple language for non-technical readers.

Objectivity – unbiased, fact-based, not opinionated.

Completeness – includes all evidence, even if not incriminating.

Integrity – properly documented chain of custody and hashes.

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Example (Simplified Report Structure)

Case Title: Unauthorized Access Investigation

Investigator: John Doe

Date: 10-09-2025

Summary:
A forensic investigation was conducted on a laptop suspected of unauthorized access.

Findings:
- Evidence of malware in C:\Windows\System32.
- Suspicious logins on 05-09-2025 from IP 192.168.1.15.
- Deleted emails recovered relating to financial fraud.

Conclusion:
The evidence strongly suggests deliberate unauthorized access and data theft.

Hash Verification:
Original Disk MD5: 7d793037a0760186574b0282f2f435e7
Forensic Copy MD5: 7d793037a0760186574b0282f2f435e7

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